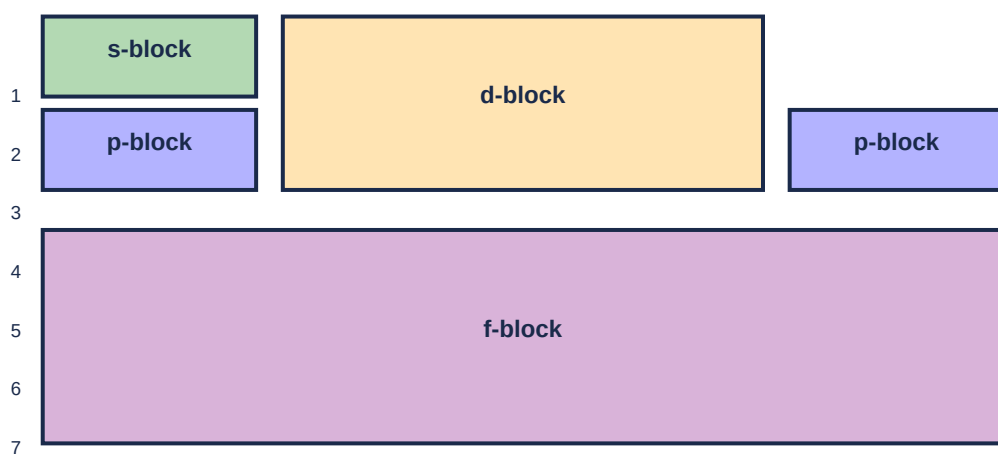


Unit 4: Periodic Classification of Elements

Complete Study Notes – Grade 9 Chemistry

Modern Periodic Table – Block Layout



Elements classified by subshell filling: s, p, d, f

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Introduction

In the nineteenth century, chemists had only a vague idea of atoms and molecules and did not know of the existence of electrons and protons. They devised the periodic table using their knowledge of **atomic masses**. Accurate measurements of atomic masses had already been made. Arranging elements according to their atomic masses in a periodic table seemed logical, as chemical behaviour was thought to be related to atomic mass.

4.1 Historical Development of Periodic Classification

4.1.1 Dobereiner's Triads

In 1829, German chemist **Johann Dobereiner** identified several groups of three elements that showed similarity in physical and chemical properties. In each **triad**, the atomic weight of the middle element is the arithmetic mean of the atomic weights of the other two.

Dobereiner's Triads

Middle element has average atomic mass of first and third

Li 7	Na 23	K 39
Ca 40	Sr 88	Ba 137
Cl 35.5	Br 80	I 127

Examples:

- Lithium (Li = 7), Sodium (Na = 23), Potassium (K = 39) $\rightarrow (7+39)/2 = 23$ (Na)
- Calcium (Ca = 40), Strontium (Sr = 88), Barium (Ba = 137) $\rightarrow (40+137)/2 = 88.5$ (Sr)
- Chlorine (Cl = 35.5), Bromine (Br = 80), Iodine (I = 127) $\rightarrow (35.5+127)/2 = 81.25$ (Br)

4.1.2 Newlands' Law of Octaves

In 1864, English chemist **John Newlands** arranged the 56 known elements in increasing order of atomic weight and observed that 'the properties of every eighth element are similar to that of the first one'. He compared this to the musical octave (eight notes) and called it the **Law of Octaves**.

Limitations of Newlands' classification:

- The noble gases (inert gases) were not yet discovered.
- Beyond calcium, the repetition of properties was not observed.

4.2 Mendeleev's Classification of the Elements

4.2.1 Mendeleev's Periodic Law

Dmitri Mendeleev (Russian) and **Lothar Meyer** (German) independently arranged elements in order of increasing atomic weights and observed that elements with similar properties appeared at regular intervals.

Mendeleev's Periodic Law:

The physical and chemical properties of elements are the **periodic function** of their atomic weights.

In 1871, Mendeleev published a periodic table with:

- 63 known elements (noble gases not included).
- 7 horizontal rows called **periods**.
- 8 vertical columns called **groups** (I to VIII).
- **Vacant sites** for undiscovered elements, whose properties he accurately predicted.

Mendeleev's Short Periodic Table

I	II	III	IV	V	VI	VII	VIII
H							
Li	Be	B	C	N	O	F	
Na	Mg	Al	Si	P	S	Cl	
K	Ca		Ti	V	Cr	Mn	Fe
Cu	Zn		Ge	As	Se	Br	

Elements arranged by atomic mass; gaps for undiscovered elements

Merits of Mendeleev's Periodic Table:

- Made the study of elements more systematic and easier.
- Predicted the existence and properties of undiscovered elements (e.g., gallium, scandium, germanium).

Demerits of Mendeleev's Periodic Table:

- Some elements were placed incorrectly despite their atomic weights (e.g., Ar (40) placed before K (39); Te (127.6) before I (126.9); Co (58.9) before Ni (58.6)).
- No fixed position for hydrogen (it was placed in Group I, but also resembled halogens).

4.3 The Modern Periodic Table

4.3.1 The Periodic Law (Modern)

The modern periodic law states that the physical and chemical properties of the elements are the **periodic functions of their atomic numbers**. This was established by **Henry Moseley** in 1913, who showed that the atomic number is the fundamental property, not atomic mass.

4.3.2 Groups and Periods

Periods – horizontal rows. There are 7 periods.

- **Period 1:** 2 elements (H, He) – filling 1s orbital
- **Period 2:** 8 elements (Li to Ne) – filling 2s, 2p orbitals
- **Period 3:** 8 elements (Na to Ar) – filling 3s, 3p orbitals
- **Period 4:** 18 elements (K to Kr) – filling 4s, 3d, 4p orbitals
- **Period 5:** 18 elements (Rb to Xe) – filling 5s, 4d, 5p orbitals
- **Period 6:** 32 elements (Cs to Rn) – filling 6s, 4f, 5d, 6p orbitals
- **Period 7:** 32 elements (Fr to Og) – filling 7s, 5f, 6d, 7p orbitals (incomplete)

Groups – vertical columns. There are 18 groups.

- Group 1 (IA): Alkali metals (ns^1)
- Group 2 (IIA): Alkaline earth metals (ns^2)
- Group 17 (VIIA): Halogens (ns^2np^5)
- Group 18 (VIIIA): Noble gases (ns^2np^6)

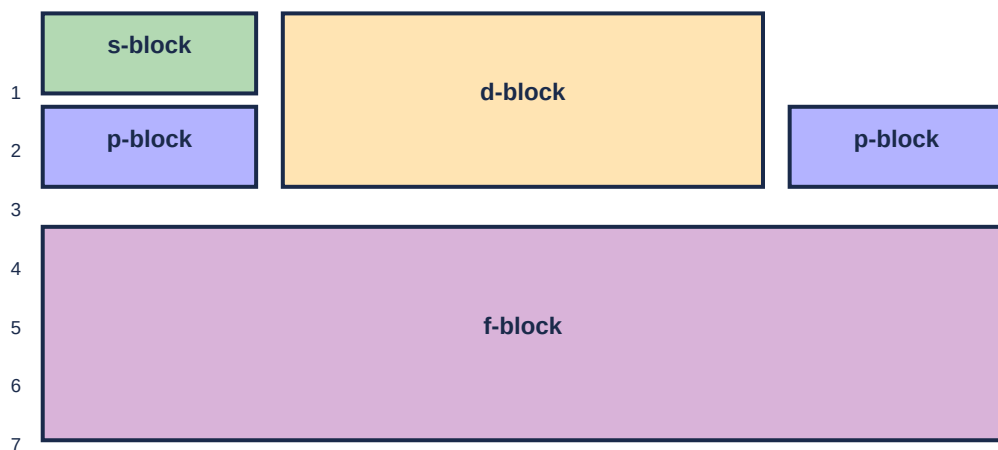
Elements in the same group have the same number of **valence electrons**, and thus similar chemical properties.

4.3.3 Classification of the Elements (s, p, d, f blocks)

Elements are classified into blocks based on the subshell in which the last electron enters.

- **s-block:** Group 1 and 2 (and He) – ns^1 or ns^2 (Alkali and alkaline earth metals)
- **p-block:** Groups 13 to 18 – ns^2np^1 to ns^2np^6 (Metals, metalloids, nonmetals; includes halogens and noble gases)
- **d-block:** Groups 3 to 12 (transition metals) – $(n-1)d^1$ to $(n-1)d^9ns^2$ (Transition metals)
- **f-block:** Lanthanides and Actinides – $(n-2)f^1$ to $(n-2)f^{14}(n-1)d^1ns^2$ (Inner transition metals)

Modern Periodic Table – Block Layout



Elements classified by subshell filling: s, p, d, f

4.3.4 The Representative Elements

The **representative elements** (main group elements) are those in Groups IA to VIIA (s-block and p-block). They have incompletely filled s or p subshells in the highest principal quantum number.

Their chemical properties are determined by the number of valence electrons.

4.4 The Major Trends in the Periodic Table

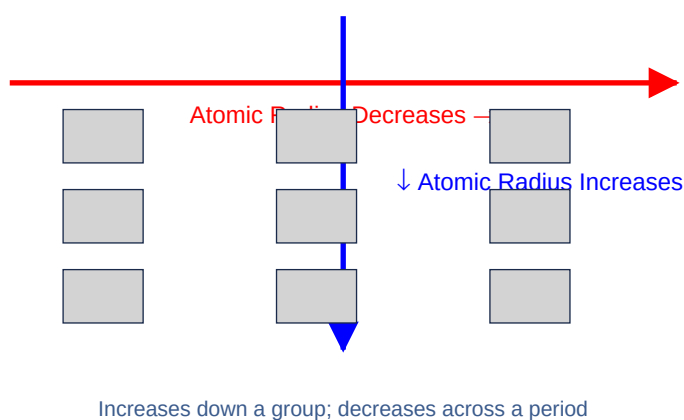
4.4.1 Atomic Radius

Atomic radius is defined as one-half the distance between the nuclei of two adjacent atoms.

Trend:

- **Across a period (left to right):** Atomic radius **decreases** because the nuclear charge increases, pulling electrons closer.
- **Down a group:** Atomic radius **increases** because new shells are added, and the outer electrons are farther from the nucleus.

Trend: Atomic Radius



Example: In Group IA, atomic radius increases from Li (152 pm) to Cs (265 pm).

4.4.2 Ionization Energy

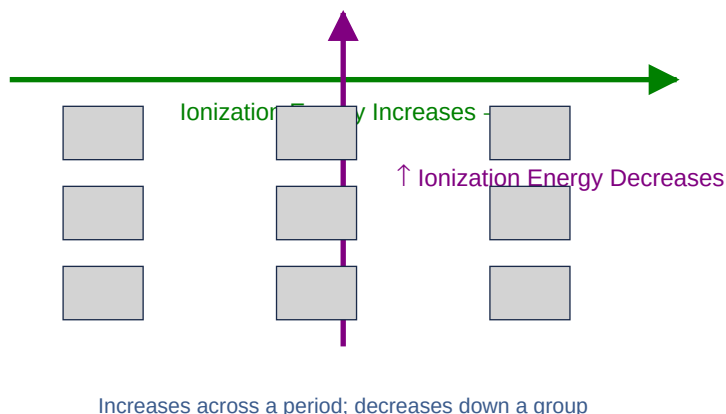
Ionization energy is the energy required to remove an electron from a gaseous atom in its ground state.

Unit: kJ/mol.

Trend:

- **Across a period:** Ionization energy **increases** because the nuclear charge increases, making it harder to remove an electron.
- **Down a group:** Ionization energy **decreases** because the outer electrons are farther from the nucleus and more shielded.

Trend: Ionization Energy



Example: First ionization energy increases from Li (520 kJ/mol) to F (1680 kJ/mol) across period 2.

Exception: Group IIA (alkaline earth metals) have higher ionization energy than Group IIIA elements due to full s-subshell.

4.4.3 Electron Affinity

Electron affinity is the change in energy when an electron is added to a neutral atom in the gaseous phase.

It is usually negative because energy is released when an atom gains an electron.

Trend:

- **Across a period:** Electron affinity becomes more negative (energy released increases) because atoms more readily gain electrons.
- **Down a group:** Electron affinity becomes less negative (energy released decreases) because the added electron is farther from the nucleus.

Example: Halogens have the most negative electron affinities (Cl: -349 kJ/mol, F: -328 kJ/mol).

Note: Fluorine has a slightly lower electron affinity than chlorine due to electron-electron repulsion in the small 2p subshell.

4.4.4 Electronegativity

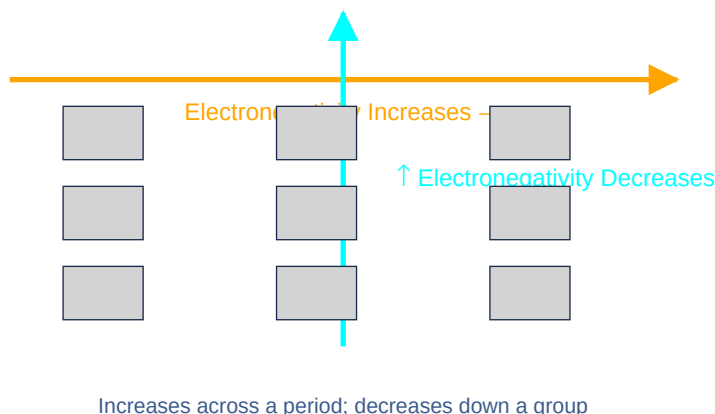
Electronegativity is the tendency of an atom in a molecule to attract the shared pair of electrons towards itself. It is a dimensionless quantity.

The most commonly used scale is the **Pauling scale**, where fluorine is assigned the highest value (4.0).

Trend:

- **Across a period:** Electronegativity **increases** because of greater nuclear charge and smaller atomic size.
- **Down a group:** Electronegativity **decreases** because the atoms are larger and the outer electrons are more shielded.

Trend: Electronegativity



Example: Electronegativity values: Li (1.0), Be (1.5), B (2.0), C (2.5), N (3.0), O (3.5), F (4.0).

Metals generally have low electronegativity (<2.0), while nonmetals have high electronegativity.

Unit Summary

- **Dobereiner's triads** – groups of three elements where the middle has the average atomic mass.
- **Newlands' law of octaves** – every eighth element has similar properties (worked for first few periods).
- **Mendeleev's periodic law** – properties are periodic functions of atomic mass; left gaps for undiscovered elements.
- **Modern periodic law** – properties are periodic functions of atomic number (Moseley).
- **Periods** – 7 horizontal rows; elements in same period have same number of shells.
- **Groups** – 18 vertical columns; elements in same group have same number of valence electrons.
- **s-block** – Groups 1 and 2 (and He); valence electrons in s orbital.
- **p-block** – Groups 13 to 18; valence electrons in p orbital.
- **d-block** – Groups 3 to 12 (transition metals); valence electrons in d orbital.
- **f-block** – Lanthanides and Actinides; valence electrons in f orbital.
- **Atomic radius** – decreases across a period, increases down a group.
- **Ionization energy** – increases across a period, decreases down a group.
- **Electron affinity** – becomes more negative across a period, less negative down a group.
- **Electronegativity** – increases across a period, decreases down a group (highest is F).

Key Terms

Atomic radius – one-half the distance between the nuclei of two adjacent atoms.

Electron affinity – energy change when an electron is added to a gaseous atom.

Electronegativity – tendency of an atom to attract shared electrons.

Group – vertical column in the periodic table.

Ionization energy – energy required to remove an electron from a gaseous atom.

Law of octaves – Newlands' observation that properties repeat every eighth element.

Mendeleev's periodic law – properties are periodic functions of atomic mass.

Modern periodic law – properties are periodic functions of atomic number.

Period – horizontal row in the periodic table.

Triad – group of three elements with similar properties, middle has average atomic mass.

End of Unit 4 – Periodic Classification of Elements

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